

| UCE617: TRAFFIC ENGINEERING AND GEOMETRIC DESIGN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |          |          |            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>L</b> | <b>T</b> | <b>P</b> | <b>Cr</b>  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>3</b> | <b>1</b> | <b>0</b> | <b>3.5</b> |
| <p><b>Course Objective:</b> The course aims to provide students with a core understanding of traffic engineering, geometric design, and Highway planning. Students will learn the core concepts of traffic flow, design control devices and road alignments, and apply statistical methods to traffic studies</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |          |          |            |
| <p><b>Core Elements of Traffic Engineering</b></p> <p><b>Traffic Flow Characteristics &amp; Studies:</b> Fundamental parameters, fundamental relationships, headways, traffic stream models, gap &amp; gap acceptance, queuing theory, volume, speed and delay studies.</p> <p><b>Capacity and Level of Service (LOS):</b> Capacity analysis of urban and rural roads, capacity of rotaries, LOS concepts as per Indian standards</p> <p><b>Traffic Control Devices:</b> Traffic signals, signal design methods (Webster's and IRC), design and placement of traffic signs, Traffic intersections, conflict points</p> <p><b>Statistical Applications:</b> Probability functions and statistics, normal distribution and applications, binomial and Poisson distributions</p> <p><b>Core Elements of Geometric Design</b></p> <p><b>Design Factors &amp; Space Standards:</b> Human and vehicle factors, space standards for highways, expressways, urban arterials, elements for cycle tracks, pedestrian walkways.</p> <p><b>Alignment Design:</b> Horizontal alignment, super-elevation, transition curve, vertical alignment, alignment coordination, consideration for hill roads, relevant IRC standards, introduction to geometric design inconsistency evaluation.</p> <p><b>Intersections:</b> Types of intersections, layout design, design considerations, channelization, on and off ramps. Rotary Intersection Design.</p> <p><b>Highway Planning Process:</b> Road alignment, surveys for highway projects (topographic, traffic, soil, and material surveys), DPR.</p> |          |          |          |            |
| <p><b>Course Learning Objectives (CLO)</b></p> <p>The students will be able to:</p> <ol style="list-style-type: none"> <li>1. Analyze traffic flow, capacity, and level of service using standard methodologies.</li> <li>2. Apply geometric design principles in highway Incorporate stakeholder engagement and community participation in decision-making processes.</li> <li>3. Design intersection and interchange design and other traffic control devices</li> <li>4. Prepare and understand a Detailed Project Report for a highway project</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |          |          |            |
| <p><b>Text Books:</b></p> <ol style="list-style-type: none"> <li>1. Khisty CJ and Lal, BK, Transportation Engineering-An Introduction, Prentice Hall, 3rd edition, 2016.</li> <li>2. R.P. Roess, E.S. Prassas and W.R. McShane, Traffic Engineering, 3rd Ed., Prentice Hall, New Jersey, 2004.</li> <li>3. C.S. Papacostas and P.D. Prevedouros, Transportation Engineering and Planning,</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |          |          |            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>3rd Ed., Prentice Hall, New Jersey, 2001. J.H. Banks, Introduction to Transportation Engineering, McGraw-Hill, New York, 2002.</p> <p>4. P.H. Wright and K. Dixon, Highway Engineering, Th. Ed., Wiley, New York, 2003. M. Meyer and E.J. Miller, Urban Transportation Planning, 2nd Ed., McGraw Hill, New York, 2001</p> <p>5. Indian Highway Capacity Manual (Indo-HCM), CRRI, New Delhi, 2017.</p> <p>6. IRC codes on geometric aspects: IRC: 38, 52, 69, 73, 86, SP-23</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Evaluation Scheme:**

| Sr. No. | Evaluation Elements                                                             | Weights (%) |
|---------|---------------------------------------------------------------------------------|-------------|
| 1.      | MST                                                                             | 30          |
| 2.      | EST                                                                             | 45          |
| 3.      | Sessionals (May include Assignments/Projects/Quizzes/Tutorials/Lab Evaluations) | 25          |